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(Key and Solutions for AIMCAT2411)

Key

SECTION – I

1. B	7. B	13. C	19. 1432
2. A	8. C	14. C	20. 2413
3. A	9. D	15. D	21. D
4. B	10. A	16. A	22. B
5. B	11. C	17. D	23. 3412
6. D	12. A	18. D	24. A

SECTION – II

1. B	6. C	11. B	16. 5
2. 77	7. B	12. D	17. 20
3. C	8. 9	13. C	18. A
4. D	9. C	14. A	19. C
5. 22	10. 15	15. C	20. A

SECTION – III

1. B	7. A	13. D	19. 10
2. 6	8. C	14. C	20. B
3. 1	9. A	15. 1	21. B
4. D	10. B	16. 5	22. C
5. 30	11. C	17. D	
6. 12	12. 6	18. A	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 513

1. Sisyphean task is an impossibly difficult task that can never really be accomplished. The author uses the term here to talk about where psychology has failed – to measure empirically 'the scales of relative valuation' of individual feelings.
Option A: The term objectification is alien to the concepts discussed here in the passage and hence hard to analyse. Even if we take it as quantification, the author doesn't talk about quantification of individual feelings as the Sisyphean task. Option A is not the answer.
Option B: This is the Sisyphean task as understood from 'The problem is that psychology has failed, according to Robbins, to measure empirically "the scales of relative valuation" of individual feelings. We can only feel an individual's subjective valuation process without being able to observe it as a fact.' Observing the individual's subjective valuation process as a fact is the impossible task that the author doesn't think we can accomplish. Hence, Option B is the answer.
Option C: The empirical measurement of scales of valuation of individual feelings is the Sisyphean task, not the empirical measurement of individual feelings directly, which the author doesn't discuss. Option C is not the answer.
Option D: It is not identifying but empirically measuring/evaluating the scales of such a process that is the Sisyphean task. Hence, Option D is not the answer. Choice (B)

2. Option A: This can be understood from the statement that 'Economics . . . is concerned with that aspect of behavior which arises from the scarcity of means to achieve given ends.' So, behaviour is, in fact, influenced by the scarcity/availability of resources/means to achieve given ends. Option A is the answer.
Option B: The analysis doesn't say whether the behaviour is adversely impacted by a scarcity of resources. Weber's analysis merely says that economics studies the behaviour arising out of the scarcity of means for achieving given ends. Option B is not the answer.

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Option C: Individual behaviour arises out of the scarcity of the resources to meet ends. It doesn't influence the ends chosen. Option C is not the answer.
Option D: Individual behaviour is not a function of just the availability of resources. It depends on the resources available to meet particular ends. Therefore, option A is a better answer than option D and option D is not the answer. Choice (A)

3. Option A: The Paretian foundation of the anti-hedonistic argument is that different possibilities offer different incentives and these incentives can be arranged in order of their intensity. This option states the opposite of that. Hence, Option A is the answer.
Option B: Even if true, this option doesn't affect the assumption that incentives can be arranged in order of their incentive. This option is irrelevant to the argument. Option B is not the answer.
Option C: Even if a single possibility offers multiple incentives, it is irrelevant to the argument about whether incentives can be compared. Option C is not the answer.
Option D: If incentives offered by various possibilities are not influenced by their scarcity, it will ***weaken the anti-hedonistic argument*** and ***not the Paretian foundation of the argument***, since the Paretian basis does not talk about scarcity of resources. This option doesn't affect the argument around whether such incentives can be compared objectively to each other and arranged in order of preference. Option D is not the answer. Choice (A)

4. Option A: The author doesn't say that there is no data to prove that an individual's valuation is rational. The author says that we can only *feel* an individual's subjective valuation process without being able to observe it as a fact. The discussion here is about the process, and how we may assess it, not how it actually is. Option A is not the answer.
Option B: The author says that while we can feel an individual's subjective valuation process, psychology has failed to measure empirically those scales, and that we won't be able to observe this process as a fact. This can be understood from "***it is really not possible to understand the concepts of choice***, of the relationship of means and ends, the central concepts of our science, ***in terms of external data***". Option B is the answer.
Option C: Consider the statement: '***Robbins minimized the importance of psychology to the simple and indisputable facts of experience***.' So, psychology's role in understanding behaviour in economics is limited to understanding or studying the facts of experience. However, whether it is the only way or not, still doesn't explain why the author called means-ends rationality psychical and not physical. Option C is not the answer.
Option D: While the author says that the structure of the means-ends relationship itself is psychical and not physical, because of the absence of observable facts, it is ambiguous and incoherent to say that there is no physical evidence—a term we cannot really define in this context. Option D is not the answer. Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 533

5. Option A: This option gives an instance of how imperial forces used local armies for their own interests and expansion. Hence, this option provides support to the author's arguments. Option A is not the answer.
Option B: This option offers a comparison between Japan and China and why one did well with respect to another. This option doesn't add value in terms of what the imperialist strategy was, according to the arguments presented by the author. Option B is the answer.
Option C: This option extends the author's argument about how imperialist forces used local military in battles, especially by arming them. Option C is not the answer.
Option D: This option is another example of how imperialists got involved in local power-plays and used that to their advantage to extend their influence. Option D is not the answer. Choice (B)

6. Consider the sentences: '***Native military support continued to be the pattern for Western imperialism in the twentieth century, whether on the offensive or on the defensive. In December 1934, an Italian force, composed largely of Somali irregulars, played the key role in a border clash at Walwal, which was used by the Italian dictator Benito Mussolini to justify the conquest of Ethiopia in 1935–36.***'
Option A: What Mussolini used as a justification for the conquest has been mentioned. Whether it was a false pretext or not has not been discussed above. Option A is not the answer.
Option B: A comparison of whether Mussolini benefited more or the Somali irregulars, has not been provided in the sentences above. Hence, Option B is not the answer.
Option C: An actual military force was involved in this case, albeit local. So, the comparison here is not between power-plays and actual military. Option C is not the answer.
Option D: This can be understood from the first sentence of the para above, that native military support, Somalis in this case, continued to help Western imperialism on the offensive and defensive. Option D is the answer. Choice (D)

7. Option A: The author explains that '***battle was an enabler of war, and war of the extension of control, but neither exhausts the subject, nor provides a sufficient explanation of success.***'

In other words, a discussion of one is incomplete without a discussion of the other, and they are intertwined. However, from this option, it is not obvious that the author wants to bring in the local power-plays alongside the military effectiveness of battles, that together fit into the larger picture of a war. Hence, Option A is not the answer.
Option B: This option explains what the author was trying to say in the last para: that talking about power-plays might undermine the military aspect of history/wars/battles, it is nevertheless important that we do discuss it. Hence, Option B is the answer.
Option C: The author in the last para doesn't really distinguish between control and military effectiveness, especially as two exclusive entities. They may be interrelated as well. Hence, Option C is not the answer.
Option D: The para doesn't really subscribe to the view that wars are all about control, or that battles are all about military. Such an assertion has not been made by the author. Option D is not the answer. Choice (B)

8. Option A: The first para actually states the contrary, that alongside the military effectiveness, one must also look at the other ignored aspect of the role of imperial forces as an outsider in local power-plays. Hence, Option A is not the answer.
Option B: The first para is not solely focused on the British exploits. It is largely about Western imperialistic forces. Hence, Option B is not the answer.
Option C: This is apt about the purpose of the first para as understood from "This is the issue of ***the extent to which the role of the outside … power has to be understood in terms of the power-plays of local interests***, especially allies, so that the conqueror is used and, indeed, becomes effective only insofar as it takes part in such relationships." The first para talks about the role of the outside conquering powers (imperialistic powers of the West) in local power struggles, and how the conquerors took part in such tussles/relationships with their local allies. Hence, Option C is the answer.
Option D: While the para casts light on how local power-plays need to be studied, it doesn't really take it to the extreme and say that only because of the local power-plays did Western imperialism become possible. Hence, Option D is not the answer. Choice (C)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 531

9. Option A: This is the first of the three facets of the framework that was discussed as understood from 'for without individuals there would be nothing that could serve as an objective of action, as an objective toward which the will could strive. ***And without objectives there could be no action…***' Option A is not the answer.
Option B: This was mentioned as the second facet as understood from 'this framework allows each one of us to appear as an individual body in a world of individuals, and ***thus as the necessary locus of agency***.' Option B is not the answer.
Option C: This is the third facet as understood from 'Third, without the efficacy of action that causality brings to the world, it would be impossible to undertake to do anything, ***to accomplish a task or act upon the world***.' Option C is not the answer.
Option D: Consider this: 'Schopenhauer, in seeking to explain why we are willing to use these structures even though they inevitably conceal reality from us, ***does not limit himself to exclusively cognitive considerations.***' So, Schopenhauer recommends going beyond the line of cognitive considerations. Option D is the answer. Choice (D)

10. Option A: The author quotes Schopenhauer by saying that since will is what we really are, and since the most basic need of the will is our most basic need, our most basic need is to be able actively to do something. That's why agency is important and everything else has to be sacrificed for it. Option A is the answer.
Option B: While this is true, it doesn't explain what the question asks, which is the conclusion of this line of thought. The conclusion is that the will's most basic need is agency, the freedom to be able to act. This option doesn't mention about agency. Hence, Option B is not the answer.
Option C: Why we veil ourselves from the world is a tangential argument that doesn't explain the conclusion of why we need to sacrifice our desires and needs. Option C is not the answer.
Option D: We don't know the causation in this case, or what comes first and what comes second. All we know is that human needs and desires do need to pave way for the will, the agency to act. Option D is not the answer. Choice (A)

11. Option A: This can be understood from '***it becomes plausible that we would be ready to subordinate and even sacrifice all other human needs and desires to its requirements for active agency.***' So, the will to act comes ahead of any other need or desire. Option A is not the answer.
Option B: Consider this: '***the fundamental structures of our experience primarily serve the most basic need of the will, which is to be able actively to will something.***' So, the need of the will is given primary importance and we impose a wrong understanding of the world because it fulfils our will, or enables our will to act. It can, therefore, be said that our understanding of the world (or our need to correctly understand the world) is not as important as the need of the will. So, this option is compatible with Schopenhauer's argumentation. Option B is not the answer.
Option C: According to Schopenhauer, our will comes before any of our other needs and desires.

It is our most fundamental need, the need to act according to our will. So, this option is incompatible with Schopenhauer's philosophy. Option C is the answer.
Option D: Since 'the most basic need of the will is our most basic need' and the most basic need of the will is the will to act, the will to act is our fundamental need. Option D is not the answer. Choice (C)

12. Option A: Consider this: 'This shift should not be surprising. It is natural to view the purportedly universal activity of structuring and distorting our own experience as serving our fundamental needs. And what one takes these needs to be depends upon one's view of the fundamental nature of human beings.' The shift here is of a priori experience enabling action rather than knowledge. But it makes sense as explained by Schopenhauer who believes that the distortion caused by a priori experience spurs action. Option A is the answer.
Option B: Our fundamental need which is the need of our will, the need to act, is served by distortion created by a priori structures. So, this option is incorrect, and not the answer.
Option C: A priori structures of experience don't serve knowledge the way they enable action according to the passage. Schopenhauer, in fact, goes beyond considerations of coherent cognition. Option C is not the answer.
Option D: A priori structures of experience may conceal reality from us, but that's not the point of the statement or the shift. Hence, Option D is not the answer. Choice (A)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 536

13. Consider this: '***But what once seemed an essential scaffold may have become a stifling straitjacket preventing physicists from reconciling quantum mechanics and gravity.***' This straitjacket is later mentioned this way: 'Unitarity dictates that a physically allowed change must correspond to a special "unitary" matrix that rotates the particle's state in Hilbert space such that the sum of the squares of its coordinates equals 1.' This is restricting because in an expanding universe this condition makes it hard to reconcile the particle states.
Option A: It is not that the Hilbert Space is hard to 'trace'. What the author means is that the available parameters to satisfy Hilbert space for an electron are not enough since there could be more possibilities for the states of the electron or any other particle. Option A is not the answer.
Option B: The number of possible states in which a particle should be present is not limiting. It is in fact ever increasing with an expanding universe, according to the author. What is limiting or has been referred to as the straitjacket is that 'Unitarity dictates that a physically allowed change must correspond to a special "unitary" matrix' and this matrix is quite limiting. Option B is not the answer.
Option C: The straitjacket refers to the restriction of possibilities for particle states, and that abstract space defining the possibilities was the earlier version of Hilbert Space. It dictated that all the coordinates must lie on a circle. With the expanding number of possibilities this restricted the physical change related to particle states. In other words, it was hard to reconcile this restriction with the reality of quantum physics. Hence, Option C is the answer.
Option D: This option doesn't explain how probabilities and amplitudes are kept in line. The probability of a particle to be in a particular degree is expressed in the form of amplitude squared. That is not clear from this option. Option D is not the answer. Choice (C)

14. Consider this: '***To visualize an isometric change, though, let the universe of this electron swell just enough to allow a third position. The electron's Hilbert space grows, but in a special way: It gains another dimension. The circle becomes a sphere, on which the particle's quantum state can swivel around to accommodate mixtures of all three locations.***' So, the suggestion was that the Hilbert Space be allowed to grow into a third dimension.
Option A: Their suggestion was not to ***expand the quantum states***. It was to expand the way the probability of actually observing a particle in a certain state can be expressed – 3D instead of 2D. Option A is not the answer.
Option B: Their argument was not that the possibilities be changed from 2D to 3D. The example of 3D (sphere) was given to help the reader understand the suggestion made by them. Their suggestion was to use isometry and enhance the number of dimensions. Hence, Option B represents one case but it is not the complete answer.
Option C: The Hilbert Space is an abstract space 'representing all possible states available to the particle'. Their suggestion was to expand this from 2D to accommodate more possibilities. This can be understood from 'Andrew Strominger and Jordan Cotler of Harvard University argue that ***a more relaxed principle called isometry can accommodate an expanding universe while still satisfying the stringent requirements [of] unitarity***'. The suggestion is that we can add more dimensions because of which "the options increase, but without unphysical consequences". So, this statement is true. Option C is the answer.
Option D: The suggestion was not to ***increase the probabilities*** but to increase the space available to represent the possible states and how it should be represented. Hence, Option D is not the answer. Choice (C)

15. Option A: From the statement 'To calculate the probability of actually observing a particle in a certain state, physicists square the amplitude which gets rid of the imaginary and negative bits and produces a positive probability. Unitarity says the sum of these probabilities must equal 1' we can understand this statement to be true. Option A is not the answer.
Option B: This can be understood from 'physicists square the amplitude which gets rid of the imaginary and negative bits and produces a positive probability.' Hence, Option B is not the answer.
Option C: All the probabilities related to a particle state must equate to 1, according to unitarity. So, indeed the probabilities have to be accounted for if unitarity has to be true. Hence, Option C is not the answer.
Option D: An amplitude is essentially the degree to which a particle is in a certain state. The amplitude doesn't control the particle's physical changes. The amplitudes change with physical changes ("As a particle's state changes, its amplitudes change too") and not the other way round. That is a misrepresentation of the information. However, 'Unitarity dictates that a physically allowed change must correspond to a special "unitary" matrix that rotates the particle's state in Hilbert space.' Unitarity is different from amplitudes. Option D is the answer. Choice (D)

16. The conflict can be summarised by this line: '***As space grows, how can the Hilbert space of possibilities not grow with it?***'
Option A: This is the presupposition. Since this statement is true, it makes it harder for the unitarity principle to be reconciled since the unitarity principle restricts the possibilities of various states of particles. Option A is the answer.
Option B: The author doesn't discuss chaos or the effects of it on particle states. So, the conflict mentioned by the author is not dependent on this presupposition. Option B is not the answer.
Option C: The conflict is with respect to options for possible states rather than for an increase in the number of particles. The author doesn't discuss about new particles. The author discusses representing possible states for particles not fitting within the conventional Hilbert Space. Option C is not the answer.
Option D: This might be true, but it doesn't explain what the conflict is that the author is pointing to. The presupposition is about the universe expanding and opening up new possibilities for particle states, which doesn't reconcile with unitarity, thus causing the conflict. Option D is not the answer. Choice (A)

Solutions for questions 17 and 18:

17. Option A: Psychological and historical processes are not designed to fulfil the function of making necessary connections. They are not ill-defined. Also, this is not the main idea of the para, because it talks about how logical propositions can be derived from only well-defined terms. Option A is not the answer.
Option B: The para talks about the paradox of defining a logical proposition from what's left in psychological and historical processes. It doesn't talk about connecting psychological or historical processes to a/any logically defined function. It is trying to talk about applying logical propositions in historical and psychological processes (which according to the author is not possible). Hence, Option B is not the answer.
Option C: A logical proposition with historical and psychological processes is a paradox, an impossibility according to the para. The para doesn't say there are logical propositions which are paradoxical, which this option does. Option C is not the answer.
Option D: This explains the essence of the para – that logical propositions need well-defined terms, since logic is all about making connections. And, that's why historical and psychological processes don't suffice. Option D is the answer. Choice (D)

18. Option A: The para doesn't talk about defining consciousness using science. Consciousness is already defined in the para. It is talking about how science is objective and consciousness is subjective, and hence, how science cannot deal with subjectivity. The option also doesn't point out that science deals with only objective phenomena. Option A is not the answer.
Option B: Science and consciousness are not fundamentally in opposition to each other. We don't have evidence to say that. All we know is that science deals with objective phenomena whereas consciousness deals with subjective phenomena. Option B is not the answer.
Option C: The para never mentions defining science and consciousness together. It talks about science being unable to deal with consciousness because the former is objective and the latter subjective. Option C is not the answer.
Option D: This option explains the essence of the para: that the phenomenon of consciousness is too subjective for science to deal with since science deals with subjectivity. Option D is the answer. Choice (D)

Solutions for questions 19 and 20:

19. Sentence 1 is a grammatically independent sentence that explains topology and how it helps understand different shapes.
Sentence 2 says that higher symmetries can detect "that" and use that to help physicists with quantum systems.
Sentence 3 refers to 'those systems' which look differently from each other and connects to a different sentence that talks about those systems.
Sentence 4 starts with a positive connector 'Similarly' and discusses quantum systems.
43 is a block because 'those systems' in 3 connects to quantum systems in 4.

'Those systems' in 3 is not the systems mentioned in 2 because in 2, the higher symmetries detect 'that', the 'that' being that these systems look completely different but obey the same rules. That's why 2 follows the 43 block. 'Similarly' in 4 connects 1 and 4 because just as topology helps focus on fundamental ways in which different shapes are the same, higher symmetries allow physicists to understand quantum systems and apply the rules to others. Hence, the answer is 1432. Ans: (1432)

20. Sentence 1 is a contextually dependent sentence (historians 'maintained' that) that talks about the theory of some historians about Operational Art. This sentence follows another sentence that offers context about Operational Art.
Sentence 2 talks about the coining of the term 'operational art'. Please note that the term was coined earlier and is only renewing interest during the 1970s and 1980s.
Sentence 3 refers to 'this position' as an extension of some argument made in a different line. We can identify that position based on the examples in 3: telegraph, rifled guns, infantry arms.
Sentence 4 has a positive connector 'also' suggesting a connection to another sentence that talks about 'Operational art'. That means 24 will be a block since 2 talks about Operational Art. Mind you, that is not the only reason for connecting 2 & 4. The 'interest' in 4 is an extension of 'received increased attention' in 2, justifying the use of 'also'.
It can also be understood that 1 and 3 are connected because 1 talks about Industrial revolution and 3 elaborates on 'this position'.
24 block precedes 13 block because 2 defines 'Operational Art'. Hence, the answer is 2413. Ans: (2413)

Solution for question 21:

21. The given bold statement starts with a connector 'In fact' that extends the idea of the previous sentence. It cannot be Blank 1 because the idea that Linkin Park's music is not dark is introduced after Blank 1 in 'Linkin Park are hardly the party commandos or death freaks you might expect.' This is followed by 'Instead, they're sensitive dudes.'
It cannot be Blank 2 because on the right side of the blank the discussion finally uses the term 'positive'.
Blank 3 is a close option but it must be understood that if the bold statement comes in Blank 3, the next sentence won't make sense. A sentence about taking positivity to the extremes cannot be followed by a sentence that talks about Linkin Park feeling the pain despite the rage and fury of the music. Only in Blank 4, the 'in fact' makes sense, as the music of Linkin Park takes positivity to its extremes, following from being positive, and feeling your pain. The band members not uttering curse words in Hybrid Theory explains the bold statement further. Hence, Option 4 is the answer. Choice (D)

Solution for question 22:

22. Option A: The causation is wrong here as the relegation of the Wars of the Roses is not purely on account of the popularity of the Hundred Years War and the Tudors. They represent interest points in Medieval and Modern History and somewhere between them, the Wars of the Roses has been left out. Option A is not the answer.
Option B: This mentions all the sub ideas of the para, that Wars of the Roses has interesting elements, and it has been denied a proper place in both Medieval and Modern History. The para also mentions a lack of interest from the historians as a reason for this neglect. Option B is the answer.
Option C: Historians have wrongly neglected The Wars of the Roses, not just in Medieval History but also in Modern History. So, the para is not just about Medieval History; it is about how Wars of the Roses didn't find a place anywhere. Option C is not the answer.
Option D: The Wars of the Roses don't find a place in Medieval or Modern history because historians ignore it in favour of other more important wars. The causation given here is wrong that because of lack of interest from political and military historians, it is relegated particularly in Medieval history. Option D is not the answer. Choice (B)

Solution for question 23:

23. Sentence 1 is an independent sentence quoting Churchill on the connection between Safety and Terror.
Sentence 2 explains how no one ('few of those responsible for maintaining the peace' meaning 'no one' wanted to rely on terror alone) was ready to rely on terror alone to keep the peace. It can be understood that 12 is a block, as 2 takes over from 1 about how 'terror' wasn't enough to keep the peace. This is further confirmed by the contrast 'However' at the beginning of this sentence that provides a departure from the idea of the first three sentences that safety came from nuclear deterrence.
Sentence 3 is an independent sentence about how and when the Nuclear Age began. It also introduces the conflict between the US and Soviet Union.
Sentence 4 explains what exactly prevented a possible war.
It can be understood that 4 is downstream to 3 because 4 talks about the consequence of the nuclear age that was discussed in 3. Similarly, the 12 block is downstream to 34 because the 12 block ends with terror not being enough reason to keep peace (the terror of nuclear exchange or the dread of a nuclear exchange preventing another total war was not enough). Hence, the answer is 3412. Ans: (3412)

Solution for question 24:

24. It cannot be Blank 2 as on either side of it the discussion is about Ben Affleck's movie. So, a sentence about the basketball shoe business won't make sense. By the same logic, it won't be Blank 3 either because the next sentence starts with 'Here' and continues talking about the movie. Between Blank 1 and Blank 4, the bold statement makes more sense in Blank 1 because then the reference in 'the boardroom hustle it took for Nike to convince a rookie named Michael Jordan to lend his name to the company's fledgling 'Air' line' becomes clear. Both the bold statement and the first sentence of the given para talk about Nike the company, and from then on, the para moves on to discuss the movie. Hence, the answer is Option 1. Choice (A)

Difficulty level wise summary - Section I

Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	5, 6, 13, 18, 19, 21, 22, 24
Difficult	1, 2, 4, 7, 8, 9, 10, 12, 14, 15, 16, 17, 23
Very Difficult	3, 11, 20

SECTION – II

Solutions for questions 1 to 5:

1. Given that each student contributed to only one category.
The students in C30, C29 must be distinct.
The students in C25 can also belong to C4 (if they went from 1st to 25th and from 27th to 30th). But this is not possible as each student contributed to only one category. Hence, the students in C25 must also be distinct.
Similarly, the students in C20, C15 must also be distinct.
The students in C10 could have contributed to C10 twice (one way being from 1st to 10th and from 12th to 21st). Hence, for C10 to be 35, there can be a minimum of 18 students.
The students in C4 could have contributed to C4 six times (one way being 1st to 4th, 6th to 9th, 11th to 14th, 16th to 19th, 21st to 24th, 26th to 29th). Hence, for C4 to be 37, there can be a minimum of 7 students.
Hence, the minimum possible number of students = 21 + 10 + 8 + 13 + 14 + 18 + 7 = 91. Choice (B)

2. The values in the categories C30 and C29 cannot be clubbed with any of the other categories. These students have to be distinct.
For C25, the students could have attended from 1st to 25th and from 27th to 30th. Hence, these 8 students in C25 will contribute to C25 and C4.
The remaining value of C4 now is 37 – 8 = 29.
For C20, the students could have attended from 1st to 20th, 22nd to 25th and 27th to 30th. Hence, these 13 students in C20 will contribute to C20 once and C4 twice. The remaining value of C4 is now 29 – 26 = 3.
For C15, one student could have attended from 1st to 15th, then from 17th to 20th, then from 22nd to 25th and then from 27th to 30th. This will account for the value of 3 in C4.
The remaining 13 students could have attended from 1st to 15th and then from 17th to 26th. Hence these 13 students contribute to C15 and to C10.
The remaining value of C10 will be 35 – 13 = 22.
The value of 22 in C10 can be accounted for by having 11 students attending from 1st to 10th and then from 12th to 21st.
Hence, the minimum number of students in the school will be 21 + 10 + 8 + 13 + 14 + 11 = 77. Ans: (77)

3. From the previous solution, the minimum number of students in the school is 77.
For C30, C29, C25 and C20, we can have the same arrangement as the above solution. For C15, consider that all the 14 students went from 1st to 15th and then from 17th to 26th. The value of C10 will become 35 – 14 = 21.
Now, C4 has a value of 3 and C10 has a value of 21.
We can have three students going from 1st to 10th, 12th to 21st and then from 23rd to 26th.
This brings down the value of C4 to 0 and C10 to 21 – 6 = 15.
We can have 7 students going from 1st to 10th and 12th to 21st.
One student can go from 1st to 10th.
In this case, the minimum number of days that a student attends school is 10 days.
The minimum cannot go below this because for C4 to have a distinct student (i.e., a student who is not combined with any other category), C10 should have one less student. C10 can only be combined with C15. Hence, the minimum value remaining in C10 can only be 35 – 14 = 21 (since C15 has a value of 14, which can contribute to C10 at most once). For this value of 21, a minimum of 11 students are required. Hence, this value cannot go below 11 and hence, C4 cannot have any distinct student.
Hence, the minimum number of days = 10. Choice (C)

4. The students can attend for 24 days in one of the following ways:
20 days and 4 days OR 10 days, 10 days and 4 days.
The maximum number of students who can attend the school on 20 days and 4 days is 13 (since C20 is 13).
The maximum number of students who can attend the school on 10 days, 10 days, and 4 days is 17 (since C10 is 35).
The remaining value of C4 is 37 – 13 – 17 = 7.

Now, one more student could have attended for four consecutive days six times during the month. Hence, the maximum number of students who attended the school for 24 days is 13 + 17 + 1 = 31. Choice (D)

5. The students can attend for 24 days in one of the following ways:
20 days and 4 days OR 10 days, 10 days and 4 days.
The minimum number of students is 77, from the above solutions.
For C25, there can be 8 students who went for 25 days and then for 4 days.
The remaining value of C4 will be 37 – 8 = 29.
For C20, assume that all the 13 students went for 20 days and then for 4 days (to maximize the number of students attending on 24 days). Then the value of C4 will be 29 – 13 = 16.
For C15, we can have 13 students attending for 15 days and then 10 days (this cannot be less as the remaining value of C10 must be 22 or less for the number of students to be 11, i.e., the minimum); one student attending for 15 days, 4 days, 4 days and 4 days.
The value of C4 will be 13 and C10 will be 22.
For C10, 11 students can attend for 10 days, 10 days and 4 days. However, in this case, C4 will have a remaining value of 2. Hence, this is not possible.
Instead, for C20, we can have two students attending for 20 days, 4 days and 4 days and 11 students attending for 20 days and 4 days. The remaining value of C4 will be 29 – 15 = 14.
For C15 and C10 we can have the same arrangement as mentioned above.
Hence, the maximum number of students who attended the school for 24 days is 11 + 11 = 22. Ans: (22)

Solutions for questions 6 to 10:

6. If a student received a score of 8 initially, it implies that according to the software he attempted 8 questions correctly and 12 questions incorrectly.
All the 8 questions that the software evaluated as correct must be incorrect.
However, it is possible that all the 12 questions that the software marked as incorrect can be correct.
Hence, the maximum possible revised score of the student is 12. Choice (C)

7. Since P, Q, R and S marked the answers to all the questions as A, B, C and D respectively, the sum of their marks, initial or revised, must be 20.
This is because the number of questions with option A, option B, option C and option D as answers must add up to 20 (even according to the erroneous key of the software).
Given that P, Q and R initially received a score of 5, 3 and 4 respectively.
Initial score of S = 20 – 5 – 3 – 4 = 8
Score of P is 5. This implies that according to the software there are 5 questions for which the correct answer is A.
However, the software considers the questions whose actual key is option D to be option A. Hence, there are actually 5 questions with option D as their answer.
Similarly, the score of Q is 3. This implies that there are 3 questions with actual answer key as option A.
The score of R is 4. This implies that there are 4 questions with actual answer key as option B.
The score of S is 8. This implies that there are 8 questions with actual answer key as option C.
The revised score of S = Number of questions with answer key as option D = 5
Required difference = 8 – 5 = 3 Choice (B)

8. Since P, Q, R and S marked the answers to all the questions as A, B, C and D respectively, the sum of their marks, initial or revised, must be 20.
This is because the number of questions with option A, option B, option C and option D as answers must add up to 20 (even according to the erroneous key of the software).
Let a, b, c and d represent the number of questions with answer key as option A, option B, option C and option D respectively.
For P, the initial marks represents the number of questions for which the answer is option D.
The revised marks represents the number of questions for which the answer is option A.
Hence, d – a = 3.
Similarly for Q, a – b = 3
For R, b – c = 1
Also, a + b + c + d = 20.
Let c = b – 1; a = b + 3; d = a + 3 = b + 6.
Hence, b + 3 + b + b – 1 + b + 6 = 20 ⇒ 4b = 12 ⇒ b = 3
Hence, a = 6; b = 3; c = 2; d = 9.
Revised score of S = 9 Ans: (9)

9. From the previous solution, option B is the correct answer to 3 questions. Choice (C)

10. From the previous solution, there are 6 questions with option A as answer; 3 questions with option B as answer; 2 questions with option C as answer; 9 questions with option D as answer.
The maximum possible initial score of the person can only be d + c = 11
In this case, the maximum possible difference can only be 11 as the revised score can be a minimum of 0.
The maximum possible revised score of a person can be a + d = 15
In this case, the maximum possible difference can be 15, if the initial scored can be 0.
We can check whether the initial score can be 0 in this case.
Since he scored 15 marks, in the first ten questions, there must be 6 questions whose answer is option A.

In the next ten questions, there must be 9 questions whose answer is option D.
To make the initial score 0, we need to check whether there are no questions with option D as the answer in the first ten questions (then the initial score will be 0 as he marked the answer as option A for the first ten questions). In the first ten questions, six questions with option A as answer, two questions with option B as answer and two questions with option C as answer.
In the next ten questions, there can be one question with option B as answer and nine questions with option D as answer.
Hence, it is possible to have a revised score of 15 and an initial score of 0.
Therefore, the maximum possible difference = 15 Ans: (15)

Solutions for questions 11 to 15:

From (ix), Dev attended MM on Day 1. Also, from (ii), Charan attended SS before Amar did. Hence, Amar could not have attended SS on Day 1. From (v) and (vi), Amar could not have attended TT or RR on Day 1. Hence, Amar must have attended OO on Day 1.

From (vii), Emran attended TT, SS and MM on three consecutive days. Also, he did not attend TT on Day 1. Hence, he could have attended them on Day 2, Day 3 and Day 4 OR Day 3, Day 4 and Day 5. However, if he attended MM on Day 5, then from (iii), Dev should have attended MM on Day 5, which is not possible. Hence, Emran must have attended TT, SS and MM on Day 2, Day 3 and Day 4. Also, Emran must have attended RR on Day 1 (he cannot attend OO on Day 1 as Amar attended OO on Day 1). Also, Emran must have attended OO on Day 5. From (iii), Dev must have attended OO on Day 3.

From (v), Charan could not have attended TT on Day 1. Hence, Binoy must have attended TT on Day 1 and Charan must have attended SS on Day 1.

Charan and Binoy must have attended OO on Day 2 and Day 4, in any order. From (viii), Charan attended TT and OO on consecutive days. If Charan attended OO on Day 2, then he must have attended TT on Day 1, which is not possible. Hence, Charan must have attended OO on Day 4 and TT on Day 3. Binoy must have attended OO on Day 2.

From (vi), Charan attended RR before Amar or Binoy and we know that Charan attended SS on Day 1, TT on Day 3 and OO on Day 4. He could not have attended RR on Day 5 for (vi) to be satisfied. Hence, Charan must have attended RR on Day 2. The only seminar remaining for Charan, which is MM, he must have attended on Day 5.

Also, Since Binoy attended OO on Day 2, he could not have attended MM on Day 2. Hence, Amar is the only person who could have attended MM on Day 2. On Day 2, Dev must have attended SS. On Day 3, Binoy must have attended MM. On Day 3, Amar must have attended RR. From (x), Binoy must have attended RR on Day 5 and Dev on Day 4. Amar must have attended TT on Day 4 and Dev must have attended TT on Day 5. Binoy must have attended SS on Day 4 and Amar must have attended SS on Day 5. From (i), SS must have been conducted by Tarun. TT and RR must have been conducted by Ravi and Varun, in any order.

	By	Day 1	Day 2	Day 3	Day 4	Day 5
MM	Umar	Dev	Amar	Binoy	Emran	Charan
SS	Tarun	Charan	Dev	Emran	Binoy	Amar
TT	Ravi/Varun	Binoy	Emran	Charan	Amar	Dev
RR	Varun/Ravi	Emran	Charan	Amar	Dev	Binoy
OO	Satish	Amar	Binoy	Dev	Charan	Emran

11. Charan attended MM on Day 5. Choice (B)

12. Dev could have attended the seminar conducted by Ravi on Day 4 (if Ravi conducted RR) or Day 5 (if Ravi conducted TT). Choice (D)

13. Dev attended SS, i.e., the seminar conducted by Tarun on Day 2. Choice (C)

14. Three persons attended SS before Binoy did. Choice (A)

15. Amar and Dev attended three seminars on consecutive days – MM, TT and RR. Choice (C)

Solutions for questions 16 to 20:

From (iii), Tarun must select two courses from OM, CM and SM.

If he selects CM and OM, he must select IDF (from (i)). Since he selected IDF, the total number of credits can be a maximum of 16. These three courses add up to

12 credits. From (iv), he has to select both OD and ME or none of them. If he selects both, the number of credits will be more than 16. Hence, he cannot select OD or ME. He can select OB to make the total number of credits 16. Hence, two cases will be possible ⇒ **(CM, OM, IDF) or (CM, OM, IDF, OB)**

If he selects SM and CM, he has to select IDF (from (i)). Also, he has to select OD, since he cannot select OM (as he has to select at least one course worth 5 credits). If he selects OD, he must also select ME (from (iv)). However, this will increase the total number of credits to 18. This will violate condition (ii). Hence, this case is not possible.

If he selects SM and OM, he has to select OB (since he has to select at least one course worth 4 credits). This is one possible case. He can select more courses in addition to these three courses. If he selects IDF, he cannot select any other course (since the total credits will be 15). If he does not select IDF, he can select both ME and OD. Hence, there are three cases possible ⇒ **(SM, OM, OB) or (SM, OM, OB, IDF) or (SM, OM, OB, OD, ME).**

16. Tarun can select the courses in five ways. Ans: (5)

17. Tarun can earn a maximum of 20 credits by selecting (SM, OM, OB, OD, ME). Ans: (20)

18. If Tarun wants to select the minimum number of courses (3), he has to select OM. Choice (A)

19. He must select an additional four courses. Choice (C)

20. He can select exactly 3 courses (CM, OM, IDF). Choice (A)

Difficulty level wise summary - Section II

Level of Difficulty	Questions
Very Easy	–
Easy	6
Medium	7, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20
Difficult	1, 8, 9, 10
Very Difficult	2, 3, 4, 5

SECTION – III

Solutions for questions 1 to 22:

1. Let us first arrange the seven numbers in the ascending order.
5, 8, 8, 9, 9, 10, 12
Now, the median will be the least possible when the remaining four numbers are all less than or equal to 5, in which case, the sixth number, i.e., 8, will be the median. Choice (B)

2. The possible combinations of placing the balls are: (1, 5, 6), (2, 4, 6), (2, 5, 5), (3, 3, 6), (3, 4, 5) and (4, 4, 4). i.e., 6 possible combinations. Ans: (6)

3. The difference must be divisible by 9, and further must be of the form 9 x m x AB (or CD) where m is between 0 to 8 (both inclusive) and AB (or CD) is any 2-digit number. Only (iii) satisfies these two conditions. Ans: (1)

4. The curve AGHIF consists of arcs AG, GH, HI and if, all of circles of radium r and central angles equal to $2\pi/3$, $\pi/3$, $\pi/3$ and $2\pi/3$.
∴ The length of the curve is equal to an arc of radius r and central angle 2π, i.e. $2\pi r$. Choice (D)

5. Let the free luggage allowance be k kgs. Let the weight of Mohan's luggage be x kg.
Weight of Sohan's luggage = $(50 - x)$ kg
Excess luggage charge would vary directly with the excess luggage Excess luggage of Mohan = $(x - k)$ kg.
Excess luggage of Sohan = $(50 - x - k)$ kg.
Excess luggage if the entire luggage belongs to one of them = $\left(50 - \frac{k}{2}\right)$ kg

$\frac{2800}{1400} = \frac{x-k}{50-x-k} \Rightarrow 3x + k = 100$ .....(1)

$\frac{2800}{6300} = \frac{x-k}{50-\frac{k}{2}} \Rightarrow 9x - 7k = 200$ ....(2)

Solving (1) and (2), $x = 30$ Ans: (30)

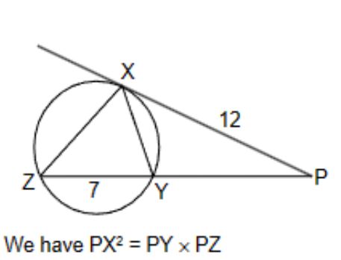
6. Time taken by P to fill ¾th of the tank = $12\left(\frac{3}{4}\right)$
= 9 minutes
Time taken by Q to empty upto ¼th of the tank from ¾th of the tank = ½ (20) = 10 minutes
Time taken for P again to fill upto ¾th of the tank from ¼th of the tank = 6 minutes.
This cycle will continue:
(9 + 10) + (6 + 10) + (6 + 10) + (6 + …), i.e., the first 57 minutes.
In the remaining three minutes, Q will empty 3/20 of the tank.
$\therefore \frac{3}{4} - \frac{3}{20} = \frac{15-3}{20} = \frac{12}{20} = \frac{3}{5}$
Hence, F = 0.6 and 20F = 12. Ans: (12)

7. $a : b = 3 : 2 = 6 : 4$
$b : c = 4 : 5$
$c : d = 5 : 6$
$d : e = 6 : 5$
From the above,
$a : b : c : d : e$ is $6 : 4 : 5 : 6 : 5$
$\therefore \frac{ad+ce}{bc+ae} = \frac{(6)(6)+(5)(5)}{(4)(5)+(6)(5)} = \frac{61}{50} = 1.22$ Choice (A)

8. $x = \log_a(3 \times 5)$
$y = \log_a(3^2 \times 5)$
$z = \log_a(3 \times 5^2)$
$10125 = 81 \times 125 = 3^4 \times 5^3$
The best way to quickly solve this question is by going through the options.

Checking for the power of 5 in choice (A)
$\frac{2x+4y+2z}{3} = \log(5^2 \times 5^4 \times 5^4)^{1/3}$
$= \log 5^{10/3}$, which is not correct.
Similarly, checking for other options, we get only for option (C)
$\frac{3x+4y+z}{3} = \frac{\log_a(3^3 \times 5^3) + \log(3^8 \times 5^4) + \log(3 \times 5^2)}{3}$
$= \frac{1}{3}\log_a(3^{12}.5^9)$
$= \log_a(3^4.5^3) = \log_a 10125$ Choice (C)

9.



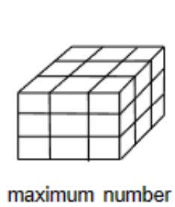
We have $PX^2 = PY \times PZ$
$\Rightarrow 144 = x(x + 7)$
$\Rightarrow x = 9$
Further, $\angle PXY = \angle XZY$ [Alternate segment Theorem]
And $\angle XPY$ is same as $\angle XPZ$
$\Rightarrow \angle PYX = \angle PXZ$
$\therefore \Delta PXY$ is similar to ΔPXZ
$\Rightarrow$ Perimeters of ΔPXY and ΔPZX will be in the ratio PX : PZ, i.e., 12 : 16 or 3 : 4
Since perimeter of PXY is 27 cm perimeter of PXZ is 36 cm. Choice (A)

10. Let the numbers of senior lecturers and junior lecturers be s and j respectively.
Using alligation, $\frac{j}{s} = \frac{250-160}{160-100} = \frac{3}{2}$
∴ 60% of the lecturers were juniors. Choice (B)

11. Let the earnings of Achyut be ₹100. Then his savings will be ₹30. Ranjit's savings will be 150% of ₹30, i.e., ₹45. If Ranjit's earnings are ₹45, then the least value of the sum of their earnings = ₹145.
Hence, the percentage mentioned will be a minimum of 145%. Choice (C)

12. For $0 < |\sin x| < 1$, we can use the formula for sum to infinite terms of a G.P
In this case $1 + |\sin x| + \sin^2 x + |\sin^3 x| + \ldots\ldots\infty$
$= \frac{1}{1-|\sin x|}$ since in 1st quadrant; $\sin x \geq 0$ and $|\sin x|$
$= \sin x$
$\Rightarrow 8^{\left[\frac{1}{1-(\sin x)}\right]} = (2^2)^3 \Rightarrow (2^3)^{\left(\frac{1}{1-\sin x}\right)} = 2^6$
Comparing powers of '2' on both sides $\frac{3}{1-\sin x} = 6$
$\Rightarrow 1 - \sin x = \frac{1}{2} \Rightarrow \sin x = \frac{1}{2} \Rightarrow x = 30° = \frac{\pi}{6}$ (radians)
$\therefore n = 6$ Ans: (6)

13.



To yield the maximum number of cubes, 2 cuts should be made in each direction, as shown above.
∴ The number of smaller cubes = 3 × 3 × 3 = 27.
Surface area of the large cube = $6a^2$.
Surface area of each small cube = $6(a/3)^2$
∴ For 27 cubes = $27 \times 6 \times (a/3)^2 = 18a^2$.
Ratio of surface areas = $6a^2 : 18a^2 = 1 : 3$
Additional amount of paint required = 2 × 3.5 = 7 litres

Alternative solution:

The cuts being parallel to the original faces, every cut results in the exposure of 2 unpainted faces of the original cube. Therefore 6 cuts will result in 12 unpainted faces being exposed, which will require 7 litres of paint. Choice (D)

14. Calculating the required values for each of the choices, it can easily be seen that k cannot be 495. Choice (C)

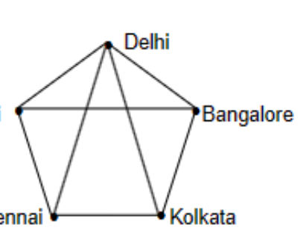
15. Let abc be the three-digit number i.e.,
$k = 100a + 10b + c$
$\Rightarrow D(k) = 99(a - c)$
Since a, b and c are in either ascending order (or) descending order $(a - c)$ cannot be 1.
So, $(a - c)$ can take values from 2 to 9. For every value of $(a - c)$, the sum of $99(a - c)$ and the number obtained by reversing $99(a - c)$ will always be equal to 1089.
$\therefore S(D(k)) = 1089$
Hence, it can take only one value. Ans: (1)

16. Since the slips are picked up one after the other without replacement, they have distinct numbers on them and 4 distinct numbers can be arranged among themselves in 4! ways. (i.e., 24 ways). Of these 24 ways only one arrangement is in ascending order.
∴ The required probability = $P = 1/24$.
Hence, $120P = 5$. Ans: (5)

17. Amount due at the end of first year = 1.1 × 50000
= ₹55,000
Amount due at the beginning of the second year = ₹45,000
Amount due at the end of the second year = 1.1 × 45000
= ₹49,500
Amount due at the beginning of the third year = ₹39,500
Amount due at the end of the third year = 1.1 × 39500
= ₹43,450
Amount due at the beginning of the fourth year = ₹33.450

Amount due at the end of the fourth year = 1.1 × 33450
= ₹36,795
Amount due at the beginning of the fifth year = ₹26,795
∴ Amount due at the end of the fifth year = 1.1 × 26795 = ₹29474.5 Choice (D)

18. The five cities represented by the vertices of a pentagon and the lines joining them represent super express ways, according to the condition given.



from the figure, the number of express ways required is 8. Choice (A)

19. If N^2 has three of its factors less than N, it has also three factors more than N. And N is also a factor of N^2.
∴ N^2 has 3 + 3 + 1 i.e., 7 factors
∴ N^2 is of the form P^6, where P is a prime number
$\Rightarrow N = P^3$
∴ $N^3 = P^9$ and P^9 has 10 factors. Ans: (10)

20. Let the cost price for Jaideep be ₹100. So, his marked price is 100(1.5) = 150 and his selling price = 150 (0.6) = 90.
∴Jaideep makes a loss of 10%. Choice (B)

21. For the runners to meet only at the poles along the track, the difference of all pairs of terms in the ratio should be a factor of 360.
By observation, in choice (B), 9 – 2 = 7 is not a factor of 360. Choice (B)

22. For the runners to meet only at the poles along the track,
- (i) the difference of all pairs of terms in the ratio or the difference of the speeds of the runners should be a factor of 360, when the runners are running in the same direction.
- (ii) the sum of the speeds of the runners should be a factor of 360 when they are running in opposite direction.

Only in choice (C), do we find that the sums as well as the differences of all pairs of terms are all factors of 360. So only in choice (C), will it not be possible for two or more runners to meet at a point on the track *between* two poles. Choice (C)

Difficulty level wise summary - Section III

Level of Difficulty	Questions
Very Easy	20
Easy	1, 7, 10, 13, 14, 18
Medium	3, 4, 5, 6, 8, 9, 11, 12, 15, 16, 17, 19
Difficult	2, 21, 22
Very Difficult	–

Done
Tip: Scroll slowly for the best results